

ClimaHealth Layer — Buildathon Dallas 2026 V1

Execution Plan

Core Objective

Build a visually polished, AI-powered healthcare operations platform that predicts COPD deterioration risk during climate events and enables proactive patient outreach before emergency escalation.

The goal is NOT to build a complete healthcare infrastructure platform.

The goal is to demonstrate:

- predictive healthcare operations
 - climate-aware patient prioritization
 - AI-generated intervention workflows
 - operational feasibility
 - enterprise-grade product thinking
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VERSION 1 SCOPE

The ONLY scenario we are building

“COPD patient deterioration prediction during Dallas heatwave / AQI spike events.”

Everything in the product should revolve around this single use case.

WHAT WE ARE ACTUALLY BUILDING

Core Product Components

1. Provider Dashboard (Main Demo Screen)

Purpose: A care coordination dashboard showing high-risk COPD patients during environmental events.

Features:

- Dallas environmental risk map
- AQI / heatwave visualization
- ranked patient queue
- risk increase percentages
- environmental trigger breakdown
- intervention approval workflow

This is the centerpiece of the demo.

2. Patient Risk Engine

Purpose: Generate believable dynamic patient risk scores.

Inputs:

- AQI
- heat index
- COPD severity
- medication adherence
- age
- smoking history
- prior utilization

Outputs:

- dynamic risk score
- risk increase percentage
- trigger explanation

IMPORTANT: This does NOT need to be a sophisticated ML system. Use lightweight scoring logic.

3. AI Outreach Generator

Purpose: Generate personalized outreach messages.

Example: "Air quality in Dallas is expected to worsen tomorrow. Please avoid prolonged outdoor activity and ensure your rescue inhaler is accessible."

Features:

- multilingual generation
- empathetic tone
- patient-specific context
- SMS preview UI

This is one of the strongest "wow factor" features.

4. Intervention Approval Workflow

Purpose: Show realistic clinical operations.

Flow:

- patient flagged
- outreach generated

- provider approves
- message scheduled
- outreach logged

The “Approve Outreach” button is critical.

5. Environmental Intelligence Layer

Purpose: Make the product visually compelling.

Data:

- Dallas AQI
- weather
- heat index
- wildfire smoke overlays

Display:

- live updating map
 - environmental severity indicators
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WHAT WE ARE NOT BUILDING

Do NOT build:

- real Epic integration
- real SMART on FHIR auth
- production HIPAA architecture
- real wearable ingestion
- mobile app
- advanced neural network training
- full multi-condition AI
- Kubernetes infrastructure
- real patient data
- FDA-grade workflows

These are time traps.

TECH STACK

Frontend

Use:

- Lovable

- Next.js
- Tailwind
- shadcn/ui
- Recharts
- Mapbox

Frontend quality matters more than backend complexity.

Backend

Use:

- FastAPI
- Python
- PostgreSQL
- lightweight APIs

Keep architecture simple.

AI Layer

Use:

- OpenAI API
- prompt-based generation
- lightweight scoring logic

Do NOT train massive healthcare models.

Data Sources

Use:

- OpenWeatherMap API
 - NOAA API
 - AQI APIs
 - synthetic patient datasets
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PRE-BUILD PLAN (BUILD BEFORE EVENT)

This should be 90% complete BEFORE arriving in Dallas.

PHASE 1 — FRONTEND DESIGN (NOW)

Build in Lovable

Pages to Create

1. Login Page

Enterprise healthcare branding.

2. Main Dashboard

Must include:

- Dallas map
- AQI indicators
- patient queue
- risk scores
- trend charts

This is the main demo screen.

3. Patient Detail View

Include:

- COPD severity
 - medication list
 - environmental exposure
 - risk timeline
 - AI explanation
 - outreach message preview
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4. Outreach Approval Screen

Include:

- generated message
 - language selector
 - approval button
 - scheduling confirmation
-

5. Analytics Page

Show:

- projected ED reduction
 - intervention counts
 - outreach metrics
 - utilization savings estimates
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PHASE 2 — SYNTHETIC DATASET

Create:

- 500 fake COPD patients
- Dallas ZIP codes
- demographic profiles
- severity levels
- medication histories
- prior exacerbation counts

Store in PostgreSQL.

PHASE 3 — RISK ENGINE

Build lightweight scoring logic.

Example:

- AQI spike = +15 risk
- severe COPD = +25 risk
- age > 70 = +10 risk
- heatwave = +20 risk

Generate:

- total risk score
- dynamic patient ranking

This is enough for the demo.

PHASE 4 — AI MESSAGE GENERATION

Use OpenAI API.

Inputs:

- patient risk
- AQI severity
- COPD severity
- preferred language

Outputs:

- empathetic outreach
- preventive recommendations

This should feel polished and human.

PHASE 5 — LIVE DEMO FLOW

Build a scripted event simulation.

Sequence: 1. AQI rises in Dallas 2. patient risk recalculates 3. patient queue reprioritizes 4. outreach generated 5. provider approves 6. message scheduled

This is your entire demo narrative.

WHAT TO BUILD DURING THE EVENT

These tasks satisfy Buildathon's "live build" expectations.

EVENT TASK 1 — DEPLOY LIVE CLOUD INFRA

Use sponsor credits for:

- cloud deployment
- API hosting
- live containers
- GPU instances if available

This demonstrates active event execution.

EVENT TASK 2 — CONNECT LIVE WEATHER APIs

Connect:

- AQI feed

- Dallas weather
- heat index
- pollution APIs

This makes the demo feel real-time.

EVENT TASK 3 — LIVE AI MESSAGE GENERATION

Use sponsor AI credits.

Generate:

- outreach messages
- multilingual communication
- intervention summaries

This is one of the strongest judging moments.

EVENT TASK 4 — LIVE EVENT STREAMING

Simulate:

- AQI spikes
- weather deterioration
- patient reprioritization

This creates “enterprise system” energy.

DEMO SCRIPT

Opening

“Healthcare reacts AFTER crises happen.”

Problem

“Climate events silently worsen chronic disease before hospitals can respond.”

Solution

“ClimaHealth predicts which patients are likely to deteriorate 48–72 hours before emergency escalation.”

Demo

Show:

- Dallas AQI event
- patient reprioritization
- AI-generated outreach
- approval workflow

Spend most of the presentation here.

Business Model

Target:

- Medicare Advantage
- ACOs
- risk-bearing healthcare groups

Revenue: PMPM enterprise SaaS.

Vision

"We are building the climate intelligence layer for healthcare operations."

MOST IMPORTANT RULES

RULE 1

Polished UI > sophisticated backend.

RULE 2

Operational realism > technical complexity.

RULE 3

One disease workflow only.

RULE 4

Focus on the demo experience.

RULE 5

Build the story, not the infrastructure.

WHAT WILL MAKE THIS WIN

Judges should leave thinking: "This feels like the future of healthcare."

That is the goal.

Not: "This is technically complete."

SUCCESS METRICS

By event day you should have:

- fully polished frontend
- live dashboard
- patient simulation
- AI outreach generation
- event-driven reprioritization
- clean demo narrative
- stable deployment

If all those are achieved, this project can realistically place very highly or win Buildathon Dallas 2026.